# Oracle 19c Data Guard with Active DataGuard

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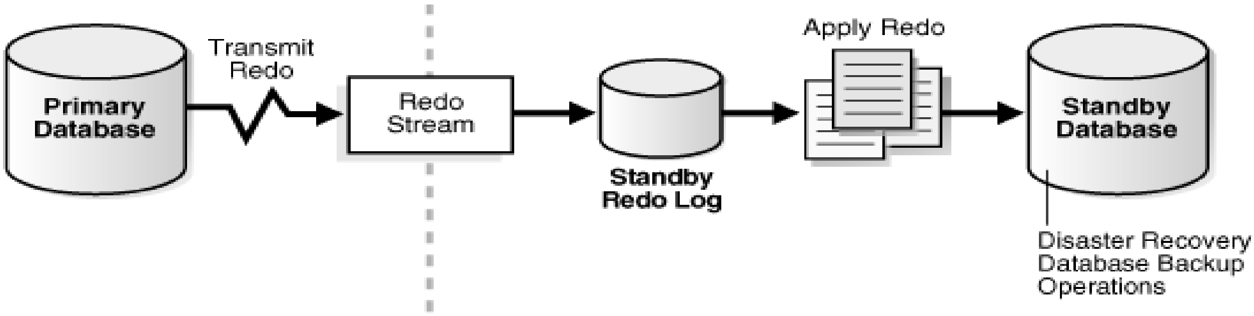
**Topic:** End-to-End Oracle 19c Physical Standby Configuration with Active Data Guard

## 1. Environment Architecture




| Title | Primary Environment | Standby Environment |
|---|---|---|
| **DB_NAME** | prod | prod |
| **SID** | prod | sec |
| **DB_UNIQUE_NAME** | prod | sec |
| **IP Address** | 192.168.0.101 | 192.168.0.102 |
| **Database Version** | 19.3.0.0 | 19.3.0.0 |
| **Hostname** | primarydb | secondarydb |
| **OS** | Oracle Linux 7 | Oracle Linux 7 |

## 2. Primary Server Configuration (IN Primary)

### Network and Hostname Setup

Configure the network and set the hostname on the primary server:

```
Bash
nmtui
# Set IP to 192.168.0.101
systemctl restart network
ifconfig
hostnamectl status
hostnamectl set-hostname primarydb
```

```
# Re-login and verify the new hostname
hostnamectl
```

Update the /etc/hosts file. It must contain a fully qualified name for the server. Both servers must have the identical entry:

```bash
vim /etc/hosts

# Add the following lines:
192.168.0.101   primarydb        primarydb
192.168.0.102   secondarydb      secondarydb
```

**Automatic & Additional OS Setup**

Install the pre-installation package and configure OS security/firewall:

```bash
# Install prerequisites
yum install -y oracle-database-preinstall-19c

# Set oracle user password
passwd oracle

# Set SELinux to Permissive
vim /etc/selinux/config
# Ensure this line exists: SELINUX=permissive
setenforce Permissive

# Disable Firewalld
systemctl stop firewalld
systemctl disable firewalld
```

**Directory Creation & Permissions**

```bash
mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
mkdir -p /u02/oradata
chown -R oracle:oinstall /u01 /u02
chmod -R 775 /u01 /u02
mkdir /home/oracle/scripts
```

**Environment Variables & Scripts**

Create the environment file setEnv.sh:

```bash
cat > /home/oracle/scripts/setEnv.sh <<EOF
# Oracle Settings
export TMP=/tmp
export TMPDIR=\$TMP

export ORACLE_HOSTNAME=primarydb
export ORACLE_UNQNAME=prod
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=\$ORACLE_BASE/product/19.0.0/dbhome_1
export ORA_INVENTORY=/u01/app/oraInventory
export ORACLE_SID=prod
export DATA_DIR=/u02/oradata

export PATH=/usr/sbin:/usr/local/bin:\$PATH
export PATH=\$ORACLE_HOME/bin:\$PATH

export LD_LIBRARY_PATH=\$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=\$ORACLE_HOME/jlib:\$ORACLE_HOME/rdbms/jlib
EOF
```

Add the environment file to the Oracle user's profile:

```bash
echo ". /home/oracle/scripts/setEnv.sh" >> /home/oracle/.bash_profile
```

Create startup and shutdown scripts:

```bash
cat > /home/oracle/scripts/start_all.sh <<EOF
#!/bin/bash
. /home/oracle/scripts/setEnv.sh

export ORAENV_ASK=NO
. oraenv
export ORAENV_ASK=YES

dbstart \$ORACLE_HOME
EOF

cat > /home/oracle/scripts/stop_all.sh <<EOF
```

```
#!/bin/bash
. /home/oracle/scripts/setEnv.sh

export ORAENV_ASK=NO
. oraenv
export ORAENV_ASK=YES

dbshut \$ORACLE_HOME
EOF

chown -R oracle:oinstall /home/oracle/scripts
chmod u+x /home/oracle/scripts/*.sh
```

**3. Installation & Database Creation (Primary)**

**Software Installation**

Log in as the oracle user and run the installer:

```
Bash
env | grep ORA
cd $ORACLE_HOME
unzip -oq LINUX.X64_193000_db_home.zip

# You can run Interactive mode:
# ./runInstaller

# OR Silent mode (Software Only):
./runInstaller -ignorePrereq -waitforcompletion -silent \
    -responseFile ${ORACLE_HOME}/install/response/db_install.rsp \
    oracle.install.option=INSTALL_DB_SWONLY \
    ORACLE_HOSTNAME=${ORACLE_HOSTNAME} \
    UNIX_GROUP_NAME=oinstall \
    INVENTORY_LOCATION=${ORA_INVENTORY} \
    SELECTED_LANGUAGES=en,en_GB \
    ORACLE_HOME=${ORACLE_HOME} \
    ORACLE_BASE=${ORACLE_BASE} \
    oracle.install.db.InstallEdition=EE \
    oracle.install.db.OSDBA_GROUP=dba \
    oracle.install.db.OSBACKUPDBA_GROUP=dba \
    oracle.install.db.OSDGDBA_GROUP=dba \
    oracle.install.db.OSKMDBA_GROUP=dba \
    oracle.install.db.OSRACDBA_GROUP=dba \
    SECURITY_UPDATES_VIA_MYORACLESUPPORT=false \
```

```
DECLINE_SECURITY_UPDATES=true
```

Run the root scripts as prompted (as root user):

```
Bash
/u01/app/oraInventory/orainstRoot.sh
/u01/app/oracle/product/19.0.0/dbhome_1/root.sh
```

**Database Creation**

Run DBCA to create the primary database:

```
Bash
dbca
```

Once installed, edit the /etc/oratab file, setting the restart flag for the instance to Y:

```
Plaintext
prod:/u01/app/oracle/product/19.0.0/dbhome_1:Y
```

## 4. Oracle Data Guard Physical Standby Configuration (Primary)

Create necessary directories:

```
Bash
mkdir -p /u02/app/oracle/arch
mkdir -p /u02/app/oracle/fast_recovery_area
mkdir -p /u02/app/oracle/rman_bkp
```

**Enable Archive Log and Force Logging**

```
SQL
SELECT log_mode FROM v$database;
ALTER SYSTEM SET log_archive_dest_1='location=/u02/app/oracle/arch' SCOPE=both;
SHUT IMMEDIATE;
STARTUP MOUNT;
ALTER DATABASE ARCHIVELOG;
ALTER DATABASE OPEN;
ARCHIVE LOG LIST;

SELECT name, force_logging FROM v$database;
ALTER DATABASE FORCE LOGGING;
SELECT name, force_logging FROM v$database;
```

**Create Standby Redo Log Files**

*Note: SRL size must match ORL size on the primary.*

```SQL
SELECT GROUP#, member FROM v$logfile;
SELECT GROUP#, THREAD#, bytes/1024/1024, MEMBERS, STATUS FROM v$log;

ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 11
'/u01/app/oracle/oradata/PROD/stb_redo11.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 12
'/u01/app/oracle/oradata/PROD/stb_redo12.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 13
'/u01/app/oracle/oradata/PROD/stb_redo13.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 14
'/u01/app/oracle/oradata/PROD/stb_redo14.log' SIZE 200M;

SELECT GROUP#, THREAD#, SEQUENCE#, ARCHIVED, STATUS FROM
V$STANDBY_LOG;
```

**System Parameter Configurations**

```SQL
SHOW PARAMETER db_name;
SHOW PARAMETER db_unique_name;
SHOW PARAMETER SERVICE_NAME;
SELECT instance_name FROM V$INSTANCE;

ALTER SYSTEM SET LOG_ARCHIVE_FORMAT='%t_%s_%r.arc' SCOPE=SPFILE;
ALTER SYSTEM SET LOG_ARCHIVE_MAX_PROCESSES=30;
ALTER SYSTEM SET REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE
SCOPE=SPFILE;
ALTER SYSTEM SET STANDBY_FILE_MANAGEMENT='AUTO';

-- Flashback Settings
SHOW PARAMETER recovery;
ALTER SYSTEM SET db_recovery_file_dest_size=20g;
ALTER SYSTEM SET db_recovery_file_dest='/u02/app/oracle/fast_recovery_area';
ALTER DATABASE FLASHBACK ON;
```

**Network Configuration (Primary)**

Configure tnsnames.ora:

```Bash
vim $ORACLE_HOME/network/admin/tnsnames.ora
```

```Plaintext
```

```
prod =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = primarydb)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = prod)
    )
  )
sec =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = secondarydb)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = sec)
    )
  )

LISTENER_PROD =
  (ADDRESS = (PROTOCOL = TCP)(HOST = primarydb)(PORT = 1521))
```

Configure listener.ora:

```
Bash
vim $ORACLE_HOME/network/admin/listener.ora
Plaintext
LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = TCP)(HOST = primarydb)(PORT = 1521))
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
    )
  )
SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = prod)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = prod)
    )
    (SID_DESC =
      (GLOBAL_DBNAME = prod_DGMGRL)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = prod)
```

```
  )
 )

ADR_BASE_LISTENER = /u01/app/oracle
```

Restart listener and copy to standby:

```
Bash
lsnrctl stop
lsnrctl start
cd $ORACLE_HOME/network/admin/
scp tnsnames.ora listener.ora oracle@secondarydb:$ORACLE_HOME/network/admin/
```

**Redo Transport Configuration**

```
SQL
ALTER SYSTEM SET log_archive_dest_2 = 'service=sec async
valid_for=(online_logfiles,primary_role) db_unique_name=sec';
ALTER SYSTEM SET fal_server = 'sec';
ALTER SYSTEM SET log_archive_config = 'dg_config=(prod,sec)';
```

**Create Initialization & Password Files for Standby**

```
SQL
CREATE PFILE='/tmp/initsec.ora' FROM SPFILE;
```

```
Bash
scp /tmp/initsec.ora oracle@secondarydb:/tmp
orapwd file=$ORACLE_HOME/dbs/orapwsec password=welcome123#
scp $ORACLE_HOME/dbs/orapwsec oracle@secondarydb:$ORACLE_HOME/dbs
```

**5. Secondary Server Configuration (IN Secondary)**

Follow the same OS prerequisites steps as the primary server. Make sure /etc/hosts matches exactly.

Set hostname:

```
Bash
hostnamectl status
hostnamectl set-hostname secondarydb
```

Adjust the setEnv.sh script for the secondary server:

```
Bash
cat > /home/oracle/scripts/setEnv.sh <<EOF
# Oracle Settings
export TMP=/tmp
export TMPDIR=\$TMP

export ORACLE_HOSTNAME=secondarydb
export ORACLE_UNQNAME=sec
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=\$ORACLE_BASE/product/19.0.0/dbhome_1
export ORA_INVENTORY=/u01/app/oraInventory
export ORACLE_SID=sec
export DATA_DIR=/u02/oradata

export PATH=/usr/sbin:/usr/local/bin:\$PATH
export PATH=\$ORACLE_HOME/bin:\$PATH

export LD_LIBRARY_PATH=\$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=\$ORACLE_HOME/jlib:\$ORACLE_HOME/rdbms/jlib
EOF
```

**6. Network and Parameter Configuration (Secondary)**

Modify listener.ora and tnsnames.ora files (which were copied from primary):

```
Bash
vim $ORACLE_HOME/network/admin/tnsnames.ora
# Add/Modify:
LISTENER_SEC =
  (ADDRESS = (PROTOCOL = TCP)(HOST = secondarydb)(PORT = 1521))

vim $ORACLE_HOME/network/admin/listener.ora
# Modify HOST to secondarydb and GLOBAL_DBNAME/SID_NAME to sec.
```

Start listener and test connections:

```
Bash
lsnrctl start
tnsping prod
tnsping sec
```

**Edit the Standby PFILE**

```bash
vim /tmp/initsec.ora
```

In Vim, perform a global replace first: :%s/prod/sec/g

Then make the following specific manual changes:

```plaintext
# Change DB Name back to prod
*.db_name='prod'

# Change Log Archive Dest
*.log_archive_dest_2='service=prod async valid_for=(all_logfiles,primary_role)
db_unique_name=prod'

# Change FAL config
*.fal_server='prod'

# Change Archive Config
*.log_archive_config='dg_config=(prod,sec)'

# Change Control Files paths back to PROD directories
*.control_files='/u01/app/oracle/oradata/PROD/control01.ctl','/u01/app/oracle/oradata/PROD/
control02.ctl'

# Add conversion parameters
*.db_file_name_convert='/u01/app/oracle/oradata/PROD/','/u01/app/oracle/oradata/SEC/'
*.log_file_name_convert='/u01/app/oracle/oradata/PROD/','/u01/app/oracle/oradata/SEC/'
*.db_unique_name='sec'
```

**Create Standby Directories & Start Instance**

```bash
mkdir -p /u01/app/oracle/admin/sec/adump /u01/app/oracle/oradata/SEC/
/u02/app/oracle/fast_recovery_area /u02/app/oracle/arch

. .bash_profile
env | grep ORA
sqlplus / as sysdba
```

```sql
CREATE SPFILE FROM PFILE='/tmp/initsec.ora';
```

```
STARTUP NOMOUNT;
SHOW PARAMETER spfile;
EXIT;
```

*Note: You must exit SQLPlus and re-enter, else cloning will fail.*

```
Bash
sqlplus / as sysdba
```

```
SQL
SHOW PARAMETER db_name;       -- Should be prod
SHOW PARAMETER db_unique_name; -- Should be sec
SHOW PARAMETER service_names;  -- Should be sec
SELECT instance_name FROM v$instance; -- Should be sec
```

**7. RMAN Duplicate (From Primary)**

Execute the RMAN duplication from the primary database:

```
Bash
rman target sys@prod auxiliary sys@sec
# OR: rman target sys/password@prod auxiliary sys/password@sec
```

```
SQL
RMAN> DUPLICATE TARGET DATABASE FOR STANDBY FROM ACTIVE
DATABASE NOFILENAMECHECK;
```

*(The logs will show backup, restore, setting newnames for temp/data files, and switching to copies. Ensure it finishes without errors).*

**8. Start Apply & Validation**

**Start MRP on Secondary**

```
SQL
sqlplus / as sysdba
ALTER DATABASE OPEN READ ONLY;
-- Start MRP command
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT;

SELECT name, open_mode FROM v$database;
-- Expected output: READ ONLY WITH APPLY
```

**Status Checks (On both Primary & Standby)**

```sql
SQL
SET LINES 999;
SELECT * FROM v$dataguard_status ORDER BY timestamp;
SELECT dest_id, status, destination, error FROM v$archive_dest WHERE dest_id<=2;
```

**Troubleshooting ORA Errors (e.g., ORA-16058) on Primary:**

```sql
SQL
ALTER SYSTEM SET log_archive_dest_state_2='DEFER';
ALTER SYSTEM SET log_archive_dest_state_2='ENABLE';
SELECT dest_id, status, destination, error FROM v$archive_dest WHERE dest_id<=2;
```

**Primary Checks:**

```sql
SQL
SELECT sequence#, first_time, next_time, applied, archived FROM v$archived_log
WHERE name = 'sec' ORDER BY first_time;
SELECT STATUS, GAP_STATUS FROM V$ARCHIVE_DEST_STATUS WHERE
DEST_ID = 2;
ARCHIVE LOG LIST;
SELECT status, instance_name, database_role, protection_mode FROM v$database,
v$instance;
SELECT MAX(sequence#) FROM v$archived_log;
```

**Standby Checks:**

```sql
SQL
SELECT process, status, sequence# FROM v$managed_standby;
SELECT sequence#, applied, first_time, next_time, name filename FROM v$archived_log
ORDER BY sequence#;
SELECT status, instance_name, database_role, protection_mode FROM v$database,
v$instance;
SELECT MAX(sequence#) FROM v$archived_log;
```

**9. Startup and Shutdown Processes**

**Standard Startup Process (On Secondary)**

```sql
SQL
STARTUP MOUNT;
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT;
```

**Shutdown Process (On Secondary)**

```
SQL
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
SHUT IMMEDIATE;
```

**Enable Active DataGuard (On Secondary)**

To open the database for reporting while logs are actively applying:

```
SQL
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
ALTER DATABASE OPEN READ ONLY;
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT;
```

**Conclusion:** This setup creates a Data Guard environment with the primary and standby databases managed, providing a robust high availability solution for your Oracle databases.